Machine Learning Project

Project Guidelines and Assessment Rubric

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1. Project objective

The purpose of this project is to demonstrate a rigorous understanding of machine learning through the *independent implementation, training, and evaluation* of a model. Students shall select a dataset of personal or academic interest from Kaggle, formulate a meaningful research question, and solve it using an appropriate supervised or unsupervised learning method.

Core requirement. The selected machine learning algorithm and its essential training or optimization procedure must be implemented *from scratch*. Students must then compare their implementation against an established Python library implementation of the same method, using a consistent experimental protocol.

2. Eligible project topics

Each project must address **at least one** of the following problem families. Students should choose an algorithm suited to their dataset and clearly justify that choice.

1. **Regression (supervised learning):** linear regression, regression trees, or neural networks for continuous-target prediction.
2. **Classification (supervised learning):** classification trees, support vector machines (SVMs), neural networks, or generative classification models.
3. **Clustering (unsupervised learning):** k-means clustering for discovering groups in unlabeled data.

A dataset may contain both labeled and unlabeled information; the chosen learning task, target (if applicable), and success criteria must be explicitly stated.

3. Implementation requirements

3.1 Components to implement independently

Students must write and explain the central mathematical and computational components of their chosen model, including, as applicable:

- **Model formulation:** predictions, decision rules, distance measures, probability distributions, tree splitting criteria, or neural-network forward propagation.
- **Objective function:** e.g., mean squared error, cross-entropy, negative log-likelihood, hinge loss, within-cluster sum of squares, or an impurity criterion.
- **Training and optimization:** appropriate procedures such as gradient descent, stochastic or mini-batch gradient descent, Newton's method, maximum-likelihood estimation, backpropagation, recursive tree construction, or k-means assignment/update iterations.
- **Prediction or inference:** applying the trained model to new observations, or assigning observations to clusters.

Important clarification: Not every model uses gradient descent or Newton's method. For example, decision trees require a split-search and tree-growing algorithm; k-means requires alternating assignment and centroid updates. Students must implement the optimization or fitting procedure *appropriate to their selected model*.

3.2 Permitted use of libraries

General-purpose libraries (e.g., NumPy, pandas, Matplotlib, Seaborn) may be used for array operations, data loading, cleaning, preprocessing, visualization, and supporting numerical operations. Established machine learning libraries (e.g., scikit-learn, XGBoost, PyTorch) may be used **only for benchmarking and comparison**, not to replace the student's core model or fitting algorithm. Library-based evaluation metrics and dataset splitting utilities are permitted. Any additional external code must be acknowledged.

4. Experimental methodology

Students are expected to:

1. **Define the problem:** describe the Kaggle dataset, its provenance, relevant features, target variable (if applicable), and research objective.
2. **Prepare the data:** document missing-value treatment, encoding, scaling, and train/validation/test partitioning as relevant. Prevent data leakage.
3. **Implement and verify:** explain the mathematical formulation, code structure, optimization procedure, stopping conditions, and any numerical checks.
4. **Evaluate fairly:** compare the from-scratch implementation with a library baseline on identical data splits and appropriate metrics.
5. **Interpret findings:** discuss model behavior, limitations, computational cost, and possible improvements.

Recommended metrics include MAE, RMSE, and R^2 for regression; accuracy, precision, recall, F1, and confusion matrices for classification; and inertia, silhouette score, and cluster visualizations for clustering, when appropriate. Students should not rely on a single metric without justification.

5. Required deliverables

1. **Written report (PDF):** problem statement, dataset description, mathematical methodology, experimental setup, labeled plots and tables, comparative results, critical discussion, and references.
2. **Source code:** clearly structured, commented, reproducible Python files or notebooks containing the complete from-scratch implementation and separate benchmark experiments.
3. **Oral presentation:** a concise explanation of the problem, model, optimization method, results, and limitations, followed by questions.

Submission deadline and advance review

Mandatory deadline: Students must submit their **final written report (PDF) and complete source code no later than seven calendar days before their scheduled oral presentation**. This advance submission is required to allow the instructor to review the methodology, inspect and execute the code, and evaluate the work before the presentation. The oral presentation and subsequent questions will take place on the scheduled presentation date.

The exact submission date is therefore determined by each student's or group's presentation date.

The submission must include instructions to reproduce the results (environment/dependencies, random seeds, dataset access, and execution order). Students should provide the Kaggle dataset URL rather than redistributing data where licensing does not permit it.

6. Assessment rubric

The project will be evaluated using the following weighted criteria:

Criterion	**Weight**
Clarity, organization, and scientific quality of the report, including informative plots	20%
Correctness, readability, modularity, and transparency of the from-scratch implementation	25%
Quality of experimental design, performance comparison, interpretation, and critical analysis	25%
Clarity, structure, timing, and professionalism of the oral presentation	15%
Accuracy and depth of responses to methodological and technical questions	15%
Total	**100%**

Assessment emphasis. High marks require demonstrable ownership of the algorithm and its optimization procedure, not merely successful execution of a pre-existing estimator. A strong submission explains *why* the method works, evaluates *how well* it works, and identifies *when* it may fail.

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